# RExercise#9_Porras

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**Part A — Creating Data Frames**

## Question #1

```
# a. The table below shows the data about shoe size and height. Create a data frame. Save it as shoesiz

shoesize_data <- data.frame(
  Shoe_size = c(6.5, 9.0, 8.5, 8.5, 10.5, 7.0, 9.5, 9.0, 13.0, 7.5, 10.5, 8.5, 12.0, 10.5,
                13.0, 11.5, 8.5, 5.0, 10.0, 6.5, 7.5, 8.5, 10.5, 8.5, 10.5, 11.0, 9.0, 13.0),
  Height = c(66.0, 68.0, 64.5, 65.0, 70.0, 64.0, 70.0, 71.0, 72.0, 64.0, 74.5, 67.0, 71.0, 71.0,
             77.0, 72.0, 59.0, 62.0, 72.0, 66.0, 64.0, 67.0, 73.0, 69.0, 72.0, 70.0, 69.0, 70.0),
  Gender = c("F", "F", "F", "F", "M", "F", "F", "F", "M", "F", "M", "F", "M", "M",
             "M", "M", "F", "F", "M", "F", "F", "M", "M", "F", "M", "M", "M", "M")
)

write.csv(shoesize_data, "shoesize.csv", row.names = FALSE)
```

```
# b. Then try to import the saved csv into the R environment. Display the last 5 rows.

shoesize <- read.csv("shoesize.csv")
tail(shoesize, 5)
```

```
##    Shoe_size Height Gender
## 24       8.5     69      F
## 25      10.5     72      M
## 26      11.0     70      M
## 27       9.0     69      M
## 28      13.0     70      M
```

```
# c. Describe the data by showing the structures and its content.

str(shoesize)
```

```
## 'data.frame':    28 obs. of  3 variables:
##  $ Shoe_size: num  6.5 9 8.5 8.5 10.5 7 9.5 9 13 7.5 ...
##  $ Height   : num  66 68 64.5 65 70 64 70 71 72 64 ...
##  $ Gender   : chr  "F" "F" "F" "F" ...
```

```
head(shoesize)
```

```
##   Shoe_size Height Gender
## 1       6.5   66.0      F
## 2       9.0   68.0      F
## 3       8.5   64.5      F
## 4       8.5   65.0      F
```

```
## 5      10.5   70.0      M
## 6       7.0   64.0      F
```

```
# d. Inspect its dimensions. How many rows? How about columns?

dim(shoesize)
```

```
## [1] 28  3
```

```
nrow(shoesize)
```

```
## [1] 28
```

```
ncol(shoesize)
```

```
## [1] 3
```

```
# Answer:
# There are 28 rows and 3 columns.
```

```
# e. Inspect by using summary() . What is the difference of the summary() from str() function?

summary(shoesize)
```

```
##    Shoe_size          Height           Gender
##  Min.   : 5.000   Min.   :59.00   Length   :28
##  1st Qu.: 8.500   1st Qu.:65.75   N.unique : 2
##  Median : 9.000   Median :69.50   N.blank  : 0
##  Mean   : 9.411   Mean   :68.57   Min.nchar: 1
##  3rd Qu.:10.500   3rd Qu.:71.25   Max.nchar: 1
##  Max.   :13.000   Max.   :77.00
```

```
# Explanation:
# str() shows the overall structure like data types and preview values, while summary() gives summary s
```

```
# f. Create a subset by males and females with their corresponding shoe size and height. What its resul

males <- subset(shoesize, Gender == "M")
females <- subset(shoesize, Gender == "F")

males
```

```
##    Shoe_size Height Gender
## 5       10.5   70.0      M
## 9       13.0   72.0      M
## 11      10.5   74.5      M
## 13      12.0   71.0      M
## 14      10.5   71.0      M
## 15      13.0   77.0      M
## 16      11.5   72.0      M
## 19      10.0   72.0      M
## 22       8.5   67.0      M
## 23      10.5   73.0      M
## 25      10.5   72.0      M
## 26      11.0   70.0      M
## 27       9.0   69.0      M
## 28      13.0   70.0      M
```

```
females
```

```
##    Shoe_size Height Gender
## 1        6.5   66.0      F
## 2        9.0   68.0      F
## 3        8.5   64.5      F
## 4        8.5   65.0      F
## 6        7.0   64.0      F
## 7        9.5   70.0      F
## 8        9.0   71.0      F
## 10       7.5   64.0      F
## 12       8.5   67.0      F
## 17       8.5   59.0      F
## 18       5.0   62.0      F
## 20       6.5   66.0      F
## 21       7.5   64.0      F
## 24       8.5   69.0      F
```

```
# Answer:
# It creates two separate data frames: one for males with 14 rows and one for females with 14 rows, eac
```

```
# g. Find the mean of shoe size and height of the respondents. Write the R scripts and its result.

mean(shoesize$Shoe_size)
```

```
## [1] 9.410714
```

```
mean(shoesize$Height)
```

```
## [1] 68.57143
```

```
# h. Is there a relationship between shoe size and height? Why?

# Explanation:
# Yes, there is a positive relationship between shoe size and height. Taller people generally have larg
```

## Question #2:

```
# A nominal variable is a categorical variable without an implied order. This means that it is impossib
# Create factors for the gender.

shoesize$Gender <- factor(shoesize$Gender)
str(shoesize$Gender)
```

```
##  Factor w/ 2 levels "F","M": 1 1 1 1 2 1 1 1 2 1 ...
```

## Question #3:

```
# When R Markdown knits a document, it uses the folder containing the .Rmd file as the working director

# Explanation:
# It forces R to look for files relative to where the .Rmd file is saved. This means if you share the p
```

## Question #4:

```
# Run str(shoes) and note the type R assigned to the Gender column. Is it what you expected? Explain wh

str(shoesize)
```

```
## 'data.frame':    28 obs. of  3 variables:
##  $ Shoe_size: num  6.5 9 8.5 8.5 10.5 7 9.5 9 13 7.5 ...
##  $ Height   : num  66 68 64.5 65 70 64 70 71 72 64 ...
##  $ Gender   : Factor w/ 2 levels "F","M": 1 1 1 1 2 1 1 1 2 1 ...
```

```
# Explanation:
# Gender was assigned as character/factor, which is expected since it contains text labels ("M" and "F"
# If a numeric column like Shoe_size gets imported as character, R will treat the numbers as plain text
```

## Question #5:

```
# Why is it good practice to check BOTH head(shoes) and tail(shoes) after importing a file, instead of

# Explanation:
# head() only shows the first 6 rows, so it cannot catch issues at the bottom of the dataset.
# Using tail() helps catch problems like trailing blank rows, extra summary/total rows left in from Exc
```

## Question #6:

```
# a. Compute for the correlation

# TODO: Use cor(shoes$ShoeSize, shoes$Height) to compute the correlation
# between shoe size and height across ALL observations

cor(shoesize$Shoe_size, shoesize$Height)
```

```
## [1] 0.7766089
```

```
# b.

# TODO: Return all rows where ShoeSize is greater than or equal to 11,
# regardless of gender

shoesize[shoesize$Shoe_size >= 11, ]
```

```
##    Shoe_size Height Gender
## 9       13.0     72      M
## 13      12.0     71      M
## 15      13.0     77      M
## 16      11.5     72      M
## 26      11.0     70      M
## 28      13.0     70      M
```

## Question #7:

```
# - Compare the mean height and mean shoe size between the female and male subsets. Describe the patter
# - Does this dataset support the idea that shoe size and height are related WITHIN each gender group,

# Explanation:
# Males have a noticeably higher average height (around 71.46 in) and larger average shoe size (10.96)
# The dataset shows that height and shoe size are positively related within each gender group (around 0
```